

Financial Question Answering

From Structured Project Evidence to Reliable Question Answering for ClimInvest

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Co-funded by
the European Union

The Challenge: Finding the Right Evidence

Climate project proposals contain the information users need — risks, benefits, costs, emissions targets and implementation plans.

This evidence is often spread across **lengthy, complex documents**.



From Evidence Challenges to a Question Answering Solution

Our goal:

Make climate-project evidence easier to find, understand and compare.

- **Find.** Quickly locate the right evidence, without searching through pages of documents.
- **Understand.** Connect related information, such as risks, mitigation measures, benefits and expected outcomes.
- **Compare.** Look across projects to identify similarities, differences and recurring patterns.



Our solution:

Build a **Question Answering** system that finds the right evidence and turns it into reliable, evidence-grounded answers.

Where Question Answering Fits in ClimInvest



Within **ClimInvest**, key information about each climate project is organised in a structured way,

This structured representation is what we call the Project Data Model, or PDM.

- PDM organises the project knowledge.
- **Question Answering** can make that knowledge easier to access.

The screenshot displays the Climinvest Studio interface. On the left, a sidebar shows the 'Project snapshot' section, including a date of '1 October 2026' and a section for '[Eastbank Community]'. The main content area shows 'EXPECTED ANNUAL IMPACT' with a table containing 'Measure', 'Expected annual generation', and 'Grid emissions factor'. Overlaid on the right is a 'Recent goal' dialog box titled 'Complete'. It contains a question from 'YOU': 'what are some of the benefits of adding solar panels to your home?'. The 'ASSISTANT' response states: 'Adding solar panels to your home can offer several potential benefits:'. It lists three bullet points: 'Lower electricity bills', 'Lower emissions', and 'Less reliance on fossil fuels'. At the bottom of the dialog, there are dropdowns for 'Mode' (set to 'Research evidence') and 'Sources' (set to 'Project sources'), a text input field with the placeholder 'Ask about your report...', and a 'Send' button.



What benefits are expected?



What are the main risks of this project?



How will those risks be managed?



What emissions reductions are targeted?



How does this project compare with similar projects?

Climate Project Proposal



100+
PAGES



Different questions require different kinds of evidence

Some questions can be answered from one project; others require evidence to be combined across several projects.

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1 Single-project questions

The answer can be found within one project's documentation.

What are the main risks of this project?

What emissions reductions are targeted?



Project A

Project details

Country	Kenya
Sector	Renewable energy
Total cost (USD m)	250

Risks

Risk	Description
Regulatory risk	Changes in policy ...
Implementation risk	Delays in grid ...
Financial risk	Currency fluctuation ...

Mitigation measures

Measure	Description
Policy engagement	Work with authorities ...
Capacity building	Training and support ...
Contingency planning	Alternative suppliers ...

Emissions target

Indicator	Target
GHG reduction (tCO ₂ e per year)	500,000



Risks

- Regulatory risk – Changes in policy ...
- Implementation risk – Delays in grid ...
- Financial risk – Currency fluctuation ...



Mitigation measures

- Policy engagement – Work with authorities ...
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Emissions reductions

- GHG reduction (tCO₂e per year)
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Find



retrieve



answer

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Mitigation measures

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Emissions reductions

- GHG reduction (tCO₂e per year) 500,000

Find → retrieve → answer

2 Cross-project questions

The answer requires evidence from several projects to compare, aggregate or identify patterns.

Which projects face similar implementation risks?

How do emissions-reduction targets differ across projects?

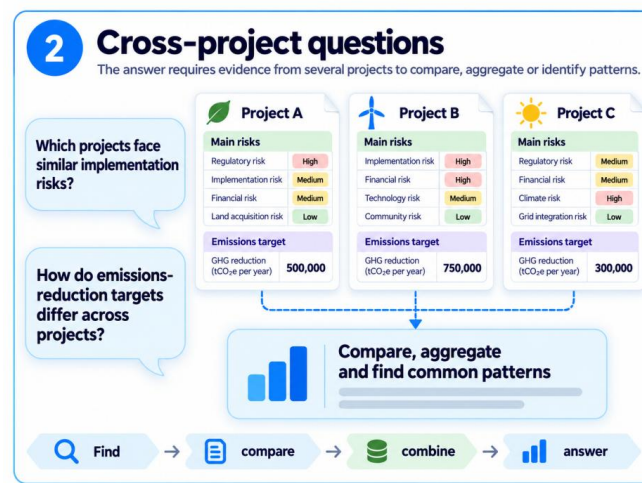
Project A	Project B	Project C
Main risks	Main risks	Main risks
Regulatory risk High	Implementation risk High	Regulatory risk Medium
Implementation risk Medium	Financial risk High	Financial risk Medium
Financial risk Medium	Technology risk Medium	Climate risk High
Land acquisition risk Low	Community risk Low	Grid integration risk Low
Emissions target	Emissions target	Emissions target
GHG reduction (tCO ₂ e per year) 500,000	GHG reduction (tCO ₂ e per year) 750,000	GHG reduction (tCO ₂ e per year) 300,000

Compare, aggregate and find common patterns

Find → compare → combine → answer

Different questions require different kinds of evidence

Some questions can be answered from one project; others require evidence to be combined across several projects.



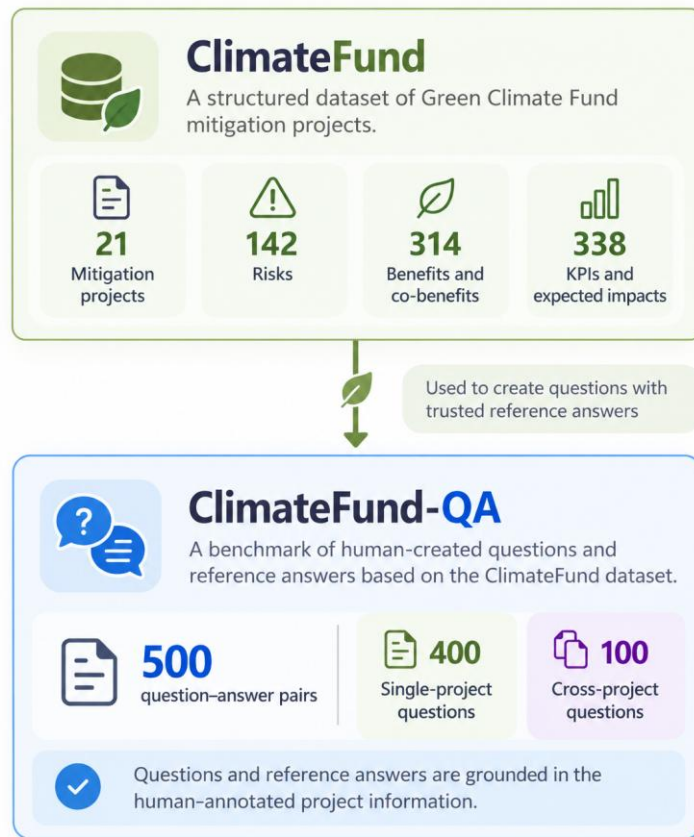
To evaluate whether AI can answer both types reliably, we need trusted questions and reference answers.

From Project Documents to reliable Question Answering

ClimateFund starts from the original Green Climate Fund project proposals.

- Human annotators identify the key project evidence, verify it against the source documents, and organise it into a consistent structure.
- This transforms information scattered across long proposals into a **structured, searchable evidence base** for analysis and Question Answering.

ClimateFund provides the trusted project evidence; **ClimateFund-QA** turns that evidence into questions and reference answers for evaluating AI systems.



Where do the trusted answers come from?

ClimateFund-QA is built from human-annotated ClimateFund evidence, grounded in the original project proposals.

Questions and reference answers are created from this verified project evidence, giving us a **trusted benchmark for evaluation**.

The AI is then evaluated against human-grounded reference answers, not answers generated by the **AI** itself.

1. Human-annotated evidence

Target cumulative emissions reduction:

15 million tCO₂eq

Information is identified from the original project proposal.



2. Human-created question

The evidence is used to formulate a clear question.

"What cumulative emissions reduction is targeted?"



3. Trusted reference answer

The expected answer is recorded before the AI is evaluated.

Reference answer:

15 million tCO₂eq

How does the AI **find an answer**?

Now that we know what the trusted answer should be, how does the AI try to find it?



This approach is called **Retrieval-Augmented Generation (RAG)**.

How do we prioritise the right evidence?

The quality of the final answer depends on whether
the system retrieves the right evidence

1 Keyword-based search (BM25)

Finds passages containing words that closely match the question.

Question

What emissions
reduction is
targeted?



Search looks for these words

emissions

reduction

target

Example matching passage



The project targets a cumulative
emissions reduction of 15 million
tCO₂eq by 2030.

2 Semantic search (E5)

Finds passages that are similar in meaning, even when they use different words.

Question

What emissions
reduction is
targeted?



Understands the same idea

expected GHG savings

cumulative avoided emissions

CO₂ mitigation objective

Example matching passage

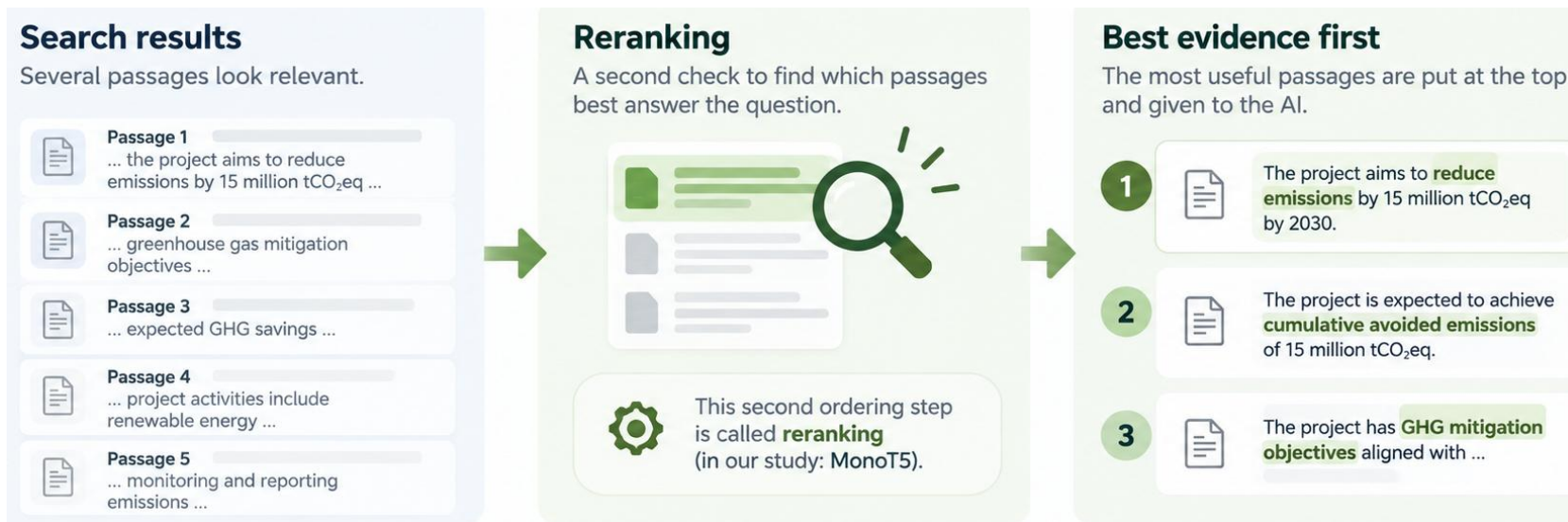


The project is expected to achieve
cumulative avoided emissions of
15 million tCO₂eq by 2030.

Once we find evidence, how do we choose the best pieces?

Search can return several passages that look relevant.

The next challenge is to put the most useful evidence first before asking the AI to answer.



Who turns the evidence into the final answer?



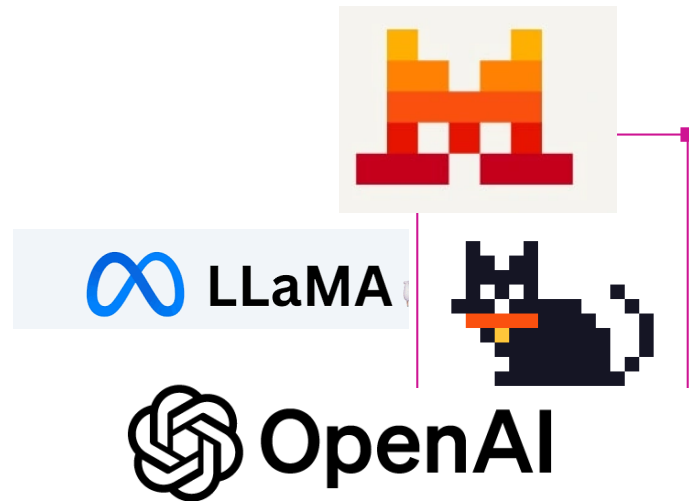
The important point is that the **AI** is not answering from its general knowledge.

We instruct it to answer using the evidence we retrieved from the project documents.

In our experiments, we tested three different models:

Mistral 7B | Llama 3.3 70B | GPT-5.4 mini

Same question + same retrieved evidence => different AI models



In Question Answering, these models are called the readers because they “read” the selected evidence and generate the answer.



How We Tested Different Ways of Answering the Same Questions



Why this matters: We ask the same questions using 12 different setups to see which combinations produce the most reliable answers.

This benchmark exposes where current QA systems still struggle.



Promising, but there is still substantial room to improve reliable QA for climate-project documents.

Questions about one project



✓ **Easier**

The system only needs to find the relevant evidence in one project document.

Best F1 score

0.279

Best semantic similarity

0.350

Questions requiring multiple projects



! **Harder**

The system must search several documents, compare information and bring it together.

Best F1 score

0.187

Best semantic similarity

0.266

Key Insights



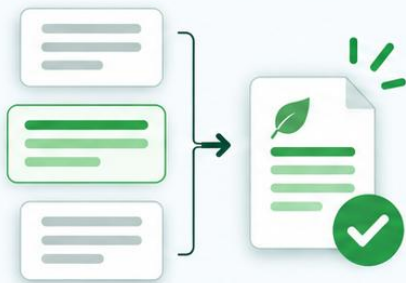
One project is easier to answer.

When the answer is contained in one project, the system has a much simpler search task.



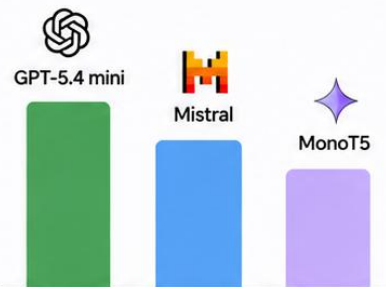
Cross-project questions are the real challenge.

The system must find evidence in several documents, compare information and bring it together into one answer.



The evidence matters as much as the AI model.

Putting the most relevant passages first generally improves the quality of the final answer.



Different AI models behave differently.

No single model performs best in every setting: GPT-5.4 mini is strongest overall for similarity in meaning, while Mistral achieves the highest single-project F1 in the reported results.

What We Have Achieved for Climinvest So Far



1 Easier access to project information

What are the main risks for this project?



- Users can ask questions instead of digging through long project documents.

2 Better support for non-expert users



- Users do not need to know exactly where the information is stored before they can find it.

3 Answers grounded in project evidence



- Questions are connected to structured information on risks, mitigation measures, benefits, KPIs and expected impacts.

4 From one project to many



- Users can explore a single project or compare evidence across several projects.

5 A way to check whether answers are reliable



- We compare AI-generated answers with trusted, human-grounded reference answers.

6 An end-to-end QA workflow



Thank you

**From complex project documents
to accessible, evidence-grounded
answers.**

ClimateFund aims to make climate-project
evidence easier to find, understand and compare.

Questions?

Miruna Clinciu

